

Zhihui Tian

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RESEARCH INTERESTS

- **AI for science and materials discovery:** physics-guided deep learning for microstructure evolution and multiscale materials modeling, learned from simulations and 3D microscopy data (Lab-DCT, TEM).
- **Graph-based learning of materials:** graph neural networks and hybrid CNN–GNN surrogates that accelerate kinetic Monte Carlo and other mesoscale simulations of 3D polycrystalline materials.
- **Generative and agentic AI for materials design:** coupling physics-based simulation surrogates with foundation models, knowledge graphs, and multi-agent reasoning for autonomous discovery.

EDUCATION

University of Florida, Gainesville, FL **May 2023 – Present**
Ph.D. in Electrical and Computer Engineering (expected Spring 2027)
Advisor: Prof. Joel B. Harley, SmartDATA Lab

University of Florida, Gainesville, FL **Aug 2021 – May 2023**
M.S. in Electrical and Computer Engineering

RESEARCH EXPERIENCE

Lawrence Livermore National Laboratory (LLNL) – Collaborative Research Project **Sep 2025 – Present**
Scalable Hybrid CNN–GNN Surrogate for 3D Grain Growth

- Led the design of a hybrid bijective autoencoder and graph neural network (AE+GNN) surrogate model for kinetic Monte Carlo grain growth, and conducted its computational experiments and benchmarking.
- Developed the latent-space inference strategy and conducted scalability studies, achieving up to 117× memory reduction and 115× runtime improvement on 3D meshes up to 160^3 relative to a GNN-only baseline.
- Established and applied physical and statistical benchmarks to evaluate surrogate predictions against Monte Carlo ground truth.

SmartDATA Lab, University of Florida – Graduate Research Assistant **May 2023 – Present**
Physics-Regulated Interpretable ML for 3D Microstructure Evolution (3D-PRIMME)

- Proposed a scalable 3D physics-guided network architecture that extrapolates from a 100^3 training domain to 1024^3 simulation domains without retraining.
- Extended the physics-regulated framework to incorporate inclination- and misorientation-dependent grain boundary dynamics.
- Applied the model to Lab-DCT experimental data and conducted quantitative comparison between ML predictions and experimental observations.

Gaussian-Sampled Mode Filter for Learning and Emulating Microstructure Evolution

- Designed Gaussian-sampled kernels to emulate grain boundary motion, mitigate lattice pinning artifacts, and provide insight into the underlying physical mechanisms.
- Demonstrated improved von Neumann–Mullins law adherence over classical Monte Carlo Potts models.
- Analyzed energy–mobility interactions for interpretable microstructure evolution modeling.

Computer Vision and Deep Learning for Experimental Microstructure Analysis

- Built and optimized a TEM image preprocessing pipeline for accurate registration of time-series micrographs.
- Adapted and fine-tuned U-Net and the Segment Anything Model (SAM) for microstructure segmentation of registered TEM images, improving segmentation accuracy and robustness.

PUBLICATIONS

Journal Articles

- [1] **Zhihui Tian**, Kang Yang, Amanda R. Krause, Michael R. Tonks, and Joel B. Harley. “3D-PRIMME: A Scalable Physics-Guided Machine Learning Framework for 3D Grain Growth.” *Materials & Design*, 2026. arxiv.org/abs/2607.04680

- [2] **Zhihui Tian**, Ethan Suwandi, Tomas Oppelstrup, Vasily V. Bulatov, Joel B. Harley, and Fei Zhou. “Scaling Kinetic Monte-Carlo Simulations of Grain Growth with Combined Convolutional and Graph Neural Networks.” *Acta Materialia*, 2026. doi.org/10.1016/j.actamat.2026.122153
- [3] Lin Yang, Vivekanand Muralikrishnan, Vishal Yadav, **Zhihui Tian**, Joel B. Harley, Amanda R. Krause, and Michael R. Tonks. “Beyond Curvature-Driven Grain Growth: Insights from Fully Anisotropic Monte Carlo Potts Simulations.” *Acta Materialia*, 2026. doi.org/10.1016/j.actamat.2025.121672
- [4] Kang Yang, Zhenhan Lin, Zekun Yang, **Zhihui Tian**, Jie Ma, José C. Príncipe, and Joel B. Harley. “Improved PCA Reconstruction-Based Unsupervised Anomaly Detection in Uncontrolled Structural Health Monitoring with Correntropy.” *IEEE Transactions on Industrial Informatics*, 2025. doi.org/10.1109/TII.2025.3584458
- [5] Kang Yang, Tianqi Liu, Zekun Yang, Yang Zhou, **Zhihui Tian**, Nam H. Kim, and Joel B. Harley. “Baseline Optimized Autoencoder-Based Unsupervised Anomaly Detection in Uncontrolled Dynamic Structural Health Monitoring.” *Structural Health Monitoring*, 2025. doi.org/10.1177/14759217251324107

Peer-Reviewed Conference Papers

- [6] **Zhihui Tian**, John Upchurch, G. Austin Simon, José Dubeux, Alina Zare, Chang Zhao, and Joel B. Harley. “Quantifying Heterogeneous Ecosystem Services with Multi-Label Soft Classification.” *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2024. doi.org/10.1109/IGARSS53475.2024.10642804
- [7] Kang Yang, Zekun Yang, **Zhihui Tian**, and Joel B. Harley. “Weight Decay Optimized Unsupervised Autoencoder-Based Anomaly Detection in Uncontrolled Dynamic Structural Health Monitoring.” *Dynamic Data Driven Applications Systems (DDDAS)*, 2024. doi.org/10.1007/978-3-031-94895-4_3

PRESENTATIONS

- [1] “3D PRIMME for Learning Grain Growth Behaviors from Simulated and Experimental Datasets.” Contributed talk, *4th World Congress on Artificial Intelligence in Materials & Manufacturing (AIM 2026)*, TMS Specialty Congress, Anaheim, CA, July 2026.
- [2] “Machine Learning of Mode Filter Grain Growth Model Characteristics.” Contributed talk, *3rd World Congress on Artificial Intelligence in Materials & Manufacturing (AIM 2025)*, TMS Specialty Congress, Anaheim, CA, June 2025.
- [3] “Quantifying Heterogeneous Ecosystem Services with Multi-Label Soft Classification.” Oral presentation, *IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2024)*, Athens, Greece, July 2024.
- [4] “Impact of Different Training Datasets on Machine Learning Based Grain Growth Model and Grain Growth Kinetics.” Contributed talk, *2nd World Congress on Artificial Intelligence in Materials & Manufacturing (AIM 2024)*, TMS Specialty Congress, Cleveland, OH, June 2024.

TEACHING

Short Course Co-Instructor, TMS 2025 Annual Meeting & Exhibition, Las Vegas, NV

Mar 2025

Grain Growth Modeling in the Fast Lane: Python, GPUs, and Machine Learning Approaches

- Half-day short course co-taught with Prof. Michael R. Tonks and Prof. Joel B. Harley; taught the mode filter grain growth model and machine learning approaches to grain growth modeling.

SKILLS

Machine Learning: graph neural networks, autoencoders, physics-informed models, surrogate modeling

Computer Vision: image segmentation (U-Net, SAM), image registration, TEM and Lab-DCT microstructure data

Materials Simulation: grain growth, Monte Carlo Potts and mode filter models, von Neumann–Mullins analysis

Programming and Tools: Python, PyTorch, PyTorch Geometric, NumPy / SciPy, MATLAB, Git, Linux / HPC

Languages: Chinese (Mandarin, native), English (fluent)